

Brief Introduction of the OpenPack Dataset and Lessons Learned from Organizing Activity Recognition Challenge Using the Dataset

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ABSTRACT

This paper presents an overview of OpenPack, a comprehensive dataset developed for recognizing packaging work activities, and discusses an activity recognition competition using this dataset. The availability of sensor datasets for recognizing work activities in industrial settings has been limited. This limitation has restricted research and development in industrial application methods based on activity recognition. To accelerate work activity recognition studies, OpenPack provides 53.8 hours of diverse sensor data, including acceleration data, keypoints, depth images, and readings from IoT devices such as handheld barcode scanners. This paper analyzes this data and propose several research directions based on OpenPack. We organized an activity recognition competition, OpenPack Challenge 2022, based on the OpenPack dataset. This paper also shares lessons learned from organizing the competition with the UbiComp and activity recognition research community. The OpenPack dataset is available at <https://open-pack.github.io/>.

CCS CONCEPTS

• **Computer systems organization** → **Neural networks**; • **Computing methodologies** → **Activity recognition and understanding**; • **Human-centered computing** → **Empirical studies in ubiquitous and mobile computing**.

KEYWORDS

Activity recognition; dataset; industrial domain

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1 INTRODUCTION

In factories and logistics centers, human workers are essential for adapting to customer and supplier demands. In 2021, Amazon employed 1.6 million people in the U.S.¹ and processed over 5 billion orders², emphasizing the need for efficient supply chains. Industry 4.0 applies digitization to streamline human work and assist decision-making. In ubiquitous computing, activity sensing and recognition for industrial workers is a significant research area [4, 6–8, 10, 12, 17, 18, 20]. However, enhancing activity recognition studies requires addressing:

- **Lack of industrial datasets:** Few datasets cover complex industrial activities like manufacturing and packing. Fig. 1 shows sensor data from a packing task, illustrating its complexity.
- **Limited modalities:** Most datasets are image-based, but occlusions in factories challenge image-only approaches.
- **Combining IoT and sensor data:** No dataset combines human movement sensor data with IoT device data, despite Industry 4.0 advancements.
- **Lack of metadata:** Public datasets lack metadata, like item combinations in packing tasks, limiting interpretation and new research designs.

This paper introduces the “OpenPack Dataset” [21, 22], a new multimodal dataset for packing task recognition in logistics centers. OpenPack collects nine sensor data types from 16 subjects, annotating 20,161 activity labels and 53,286 basic action labels. Features include: (1) **Largest multimodal activity dataset** in the industrial domain with IMUs, depth images, and LiDAR point clouds. (2) **Rich metadata** on activities and subjects, including experience level and physical characteristics, allowing diverse research design. (3) **IoT**

¹<https://www.macrotrends.net/stocks/charts/AMZN/amazon/number-of-employees>

²<https://www.seattletimes.com/business/amazon-is-upss-biggest-customer-and-biggest-competitive-threat>

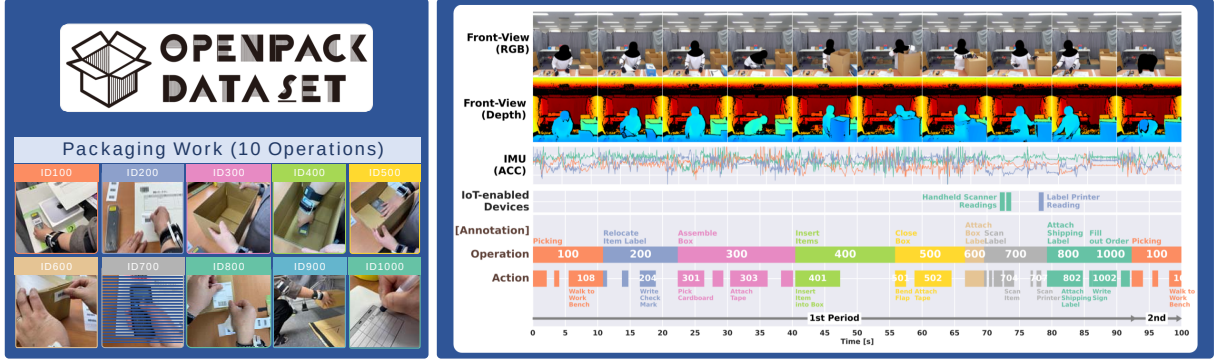


Figure 1: Example data of OpenPack dataset

device readings for complex packing task recognition, enhancing accuracy.

We analyze the dataset and propose several research directions based on OpenPack. We also organized the OpenPack Challenge 2022, an activity recognition competition at PerCom 2023³, and share insights from it with the UbiComp community.

2 RELATED WORK

Although many datasets for daily activity recognition, including multimodal, image-based, and IMU-based ones, are widely available, few exist for work activity recognition in the industrial domain. The only known dataset in logistics is LARa [9, 10], but its activity classes do not fully capture work process complexity.

The InHARD [3] and ABC Bento Packaging [1] datasets support human-robot collaboration in industrial settings. InHARD includes RGB and 3D skeleton data from 16 subjects assembling parts, while ABC Bento focuses on packing tasks and errors in bento manufacturing. These datasets lack sufficient data for deep learning algorithms and do not include accelerometer sensor data.

Complex procedural activity datasets, like CMU-MMAC [14], 50 Salads [15], and Cooking Activity Dataset [5], focus on cooking activities. Other image-based procedural activity datasets include IKEA-ASM [2], Assembly101 [13], and COIN [16]. Despite many such datasets for daily manual tasks, few exist for the industrial domain.

3 OPENPACK DATASET

The OpenPack Dataset^{4,5}, a large-scale multimodal dataset for work activity recognition in the industrial domain, includes 2048 packing tasks (work periods) performed by 16 subjects, totaling 53.8 hours of data across 104 sessions. Details are in [21, 22], and this paper provides an overview.

3.1 Packaging Work

Figure 1 shows the iterative sequence of complex operations like “assembling a cardboard box” and “packing items.” Each shipping order involves 10 operations (activities) and use of IoT devices like

handheld scanners. Variations in product size and shipping boxes make recognition challenging.

3.2 Data Collection

3.2.1 Data Collection Environment. With industrial engineers’ help, we replicated a logistics center environment (3m × 5m) with work and back tables, air cushions, and a trash can. Tools included a handheld scanner, label printer, and packing tape. Cardboard boxes were available in four sizes.

3.2.2 Participants. Sixteen participants aged 20s to 50s participated, with 12 having 1 month to 4 years of packing experience.

3.2.3 Data Modalities. OpenPack collects nine data types: acceleration, gyroscope, quaternion, Blood Volume Pulse (BVP), Electrodermal Activity (EDA), body temperature, skeletal data, LiDAR point clouds, and depth images. Four IMU units and two Empatica E4 sensors were attached to participants.

3.2.4 Data Collection Procedure. Participants received instructions, gave consent, practiced, and calibrated sensors. Each collected data in up to five sessions with breaks, performing 20 consecutive packing tasks per session. Four scenarios were used to simulate various conditions.

3.2.5 Data Collection Scenarios. Four scenarios simulated factors like procedure changes, varying product sizes, and non-standard actions. Scenario 1 followed standard instructions; Scenario 2 allowed procedure changes; Scenario 3 included non-standard actions; Scenario 4 applied pressure with an alarm sound.

3.2.6 Metadata. OpenPack provides metadata on participants (experience, dominant hand, gender, age) and shipping orders (product IDs, sizes).

3.2.7 Annotations. Experienced data labelers annotated data based on RGB images over a year, identifying 10 work activity classes and 32 basic action classes, like “pick up cardboard” and “attach tape” within “assemble box” activities, useful for logistics managers.

3.3 Dataset Analysis

3.3.1 Annotation Summary. Figure 2 shows the total length for each activity, revealing significant variation in labeled segment lengths. This variation reflects differences in the time required for

³<https://open-pack.github.io/challenge2022>

⁴<https://open-pack.github.io/>

⁵<https://zenodo.org/doi/10.5281/zenodo.5909086>

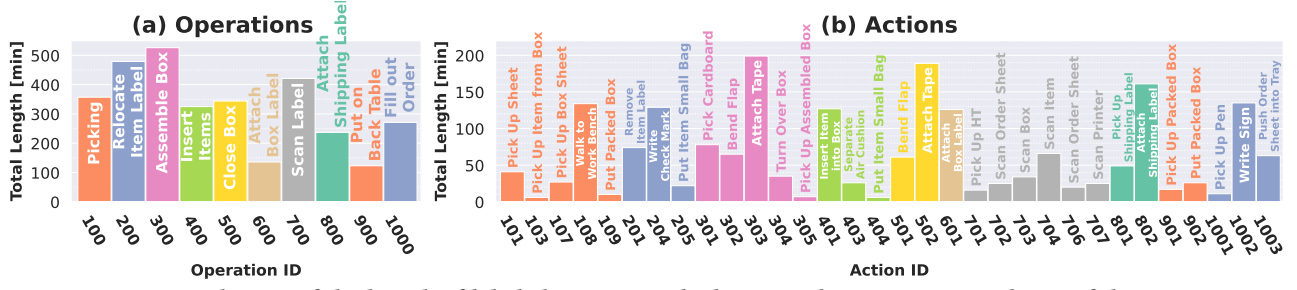


Figure 2: Distribution of the length of labeled activities. The horizontal axis represents the ID of the activities.

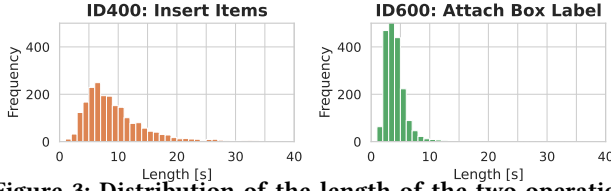


Figure 3: Distribution of the length of the two operation classes

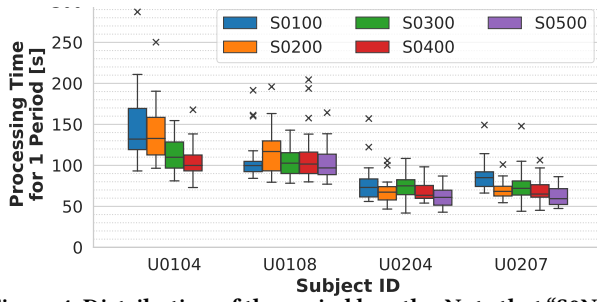


Figure 4: Distribution of the period lengths. Note that “S0N00” shows the N -th data collection session.

different work operations/actions and their frequency. Recognizing these activities in OpenPack requires addressing this class imbalance problem.

3.3.2 Variation in Operations. Figure 3 shows the distribution of the length of four operations; “insert items” and “attach box label.” Simple operations such as “attach box label,” which are not affected by the number of items in an order sheet, only exhibit small variations in length. In contrast, operations such as “insert items,” which depend on the numbers and sizes of items, exhibit long-tailed distributions. Even for the same work activities, the data exhibit a wide variation depending on different situations.

3.3.3 Variations in Period Length. Figure 4 shows the distribution of period lengths calculated for each user and session. The time required to process a single order, and there are significant differences among subjects. The working speeds of the subjects vary significantly. For example, the work speed of experienced subject U0204 is particularly fast; this worker completed a single period in 70.1 seconds on average. In contrast, the average duration for all workers is 96.4 seconds.

3.4 Research Directions

Here, we discuss future research directions using OpenPack.

- **Activity Recognition Using Metadata:** Sensor data varies based on conditions like item number. Using order information (number and size of items) managed by an online system can enhance activity recognition performance.

- **Speed-Robust Activity Recognition:** Work speed affects recognition performance. Developing techniques robust against speed variations is essential.

OpenPack’s rich metadata and annotations also support tasks like: (1) Transfer learning across sensor locations, subjects, and modalities, (2) Skill evaluation using sensor data and work experience metadata [19], (3) Counting actions and packed items [11], (4) Estimating worker fatigue using sensor and vital data, (5) Detecting mistakes in the work process [13].

4 ORGANIZING OPENPACK CHALLENGE 2022

4.1 Overview

We hosted the OpenPack Challenge 2022, a competition designed to evaluate participants’ models on their ability to recognize activities using the OpenPack dataset. Participants submit their recognition results for the provided test data through the competition management site, Codalab⁶.

4.2 Task

Participants develop models to recognize work activities that make up packing tasks from four IMU data and skeletal data, etc. Packing tasks consist of 10 activity classes, and dense labeling is performed to predict the activity class for each 1-second time step. For model training, labeled data from 11 subjects are available, while the test data comprises data from 4 subjects. Participants submit their recognition results for the test data to the Codalab site, and recognition performance is calculated as an F1 score.

4.3 Timeline

The competition started at Oct 15, 2022, and the submission deadline was Jan 15, 2023. After the deadline, the results were tallied, and the top winners were requested to write and submit reports.

4.4 Publicity Activities

To recruit participants for the competition, we conducted external publicity activities. The publicity mainly utilized academic mailing

⁶<https://codalab.lisn.upsaclay.fr/>

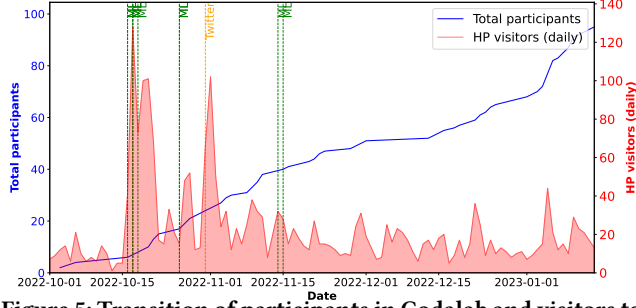


Figure 5: Transition of participants in Codalab and visitors to the OpenPack website. The vertical lines indicate the timing of recruitment posts.

lists (ML), posting to MLs related to ubiquitous mobile computing and machine learning. Furthermore, we used PerCom Twitter (now X) to announce the competition.

4.5 Awards and Ceremony for the Top Winners

This competition received financial support from the hierarchical bio-navigation⁷ and Toshiba Corporation. The hierarchical bio-navigation provided up to 750,000 JPY for travel expenses for the top 5 teams to attend the award ceremony. Toshiba Corporation provided prize money for the top 3 teams: 100,000 JPY for the first place, 40,000 JPY for the second place, and 20,000 JPY for the third place. The award ceremony took place at the 2nd International Workshop on Behavior Analysis and Recognition for Knowledge Discovery (BiRD2023), held in conjunction with PerCom 2023.

4.6 Statistical Information on the Competition

Figure 5 displays the change in the number of competition registrants alongside the timing of participant recruitment via ML and other platforms. It shows a steady increase in participants as recruitment efforts intensified. The figure also highlights the trend in visitors to the OpenPack Challenge 2022 website, with a significant correlation between the number of visitors and recruitment efforts. Additionally, a post on PerCom’s Twitter (now X) on November 1 had a significant impact. Tweets with the hashtag “#IEEEPerCom” were featured on the PerCom homepage, and the OpenPack-specific Twitter account used this hashtag for recruitment posts.

4.7 Statistical Information on the Task

Figure 6 shows the transition of the highest F1 scores among competition participants. An F1 score of around 0.90 was achieved within the first month, with gradual improvements afterward. Table 1 lists the final scores of the top winners, and the figure also shows the top winners’ F1 score transitions. Early participants ranked high, with the top 3 teams refining their models after reaching a 0.9 F1 score within a month. In the final month, the 1st place changed hands several times. These results suggest that the 3-month competition duration was reasonable.

The method of the first-place team, tomoon, involved feature extraction from multimodal data such as IMU and skeletal data

⁷<https://bio-navigation.jp/>

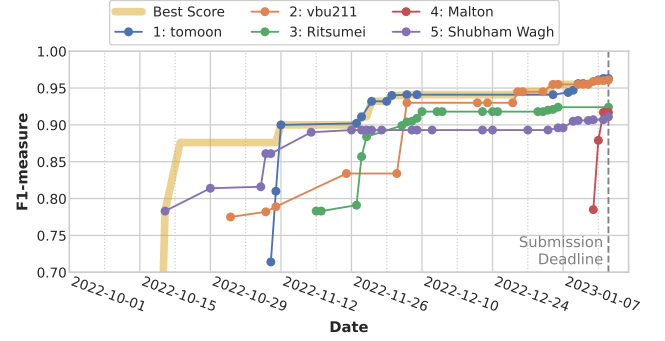


Figure 6: Transition of the highest F1 scores in the competition and among the top winners

Table 1: Top winners and their F1 scores

Rank	Team Name	F1 Score	Solution
1	tomoon	0.9633	Multi-modal
2	vbu211	0.9592	Multi-modal
3	Ritsumei	0.9241	Multi-modal
4	Malton	0.9171	IMU only
5	Shubham Wagh	0.9112	IMU only

using Transformers, and then fusing these features with another Transformer. Additionally, during training, cross entropy from classification performed on each modality was also used in the loss function. The second-place team, vbu211, extracted features from multimodal data such as IMU, skeletal data, and depth data using methods like DeepConvLSTM, then fused the features from each modality. During fusion, self-attention was applied to each modality before using the attention to weight the data from other modalities. The third-place team, Ritsumei, extracted features from multimodal data using CNNs for IMU data and Graph Convolutional Networks for skeletal data, then integrated the classification results estimated from each modality. The fourth-place team, Malton, simply classified using DeepConvLSTM with only IMU data but made adjustments such as removing data that could potentially be noise during training. The fifth-place team, Shubham Wagh, recognized activities using a network comprising CNN, Bi-LSTM, and self-attention with only IMU data. These results indicate that methods combining multimodal data occupied the top three positions. Shubham Wagh, in fifth place, submitted recognition results early but was unable to significantly improve scores in the later stages using only IMU data. The top three teams used multimodal data, but their fusion methods differed. The first and second-place teams used attention in their fusion, indicating that innovative fusion methods can lead to high recognition performance.

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